Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist, *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Launched a location intelligence consultancy specializing in retail and service sector expansion, successfully identifying $1.5M+ in untapped expansion opportunities for clients.
- Architected a revenue driver tree framework (Traffic × Capture Rate × Average Ticket Price) coupled with structural margin models to translate macro client profit objectives into precise, localized break-even targets.
- Estimated incremental revenue for net-new sites by applying advanced observational causal inference methodologies (Double ML, Causal Forests) over a 12-year county panel data framework.
- Validated causal findings via rigorous robustness checks, deploying Difference-in-Differences (DiD) setups and out-of-sample back-testing against established operational sites.
- Engineered automated Python simulation pipelines utilizing Monte Carlo risk modeling and multi-scenario (low/base/high) analysis; implemented 150+ unit tests to secure data integrity while slashing client report turnaround times by 91%.

Senior Data Scientist, *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Developed targeted uplift and audience segmentation models on millions of HIPAA-compliant healthcare records; models were adopted across 87% of active marketing campaigns, capturing an average $10K ad spend reduction per campaign.
- Designed and deployed a centralized in-house experimentation and reporting platform utilized by Data Science and Strategy teams to validate campaign lift, automate metrics monitoring, and stream data to Power BI pipelines.
- Discovered that insurance claims and customer support tickets served as primary leading indicators for user churn; successfully integrated these features into the core churn predictive model.
- Re-architected the churn modeling infrastructure on Spark and Databricks, optimizing memory distribution to cut execution runtime by 48% and reduce total memory footprint by 23%.
- Formulated a certified Gross Lead Rate (GLR) definition and data standard for a major client lacking native lead-tracking infrastructure; collaborated with data engineering to systematically date leads, map attribution, and isolate broker bias.

Founding Machine Learning Engineer, *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, engineered the search ranking and dynamic pricing engines, defining success via NDCG@10 and MAE; achieved a $420K increase in annualized gross profit.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) that cut erroneous and outlier shipping prices by 13%, reducing cost leakage while maintaining an 18% contribution margin.
- Architected automated drift monitoring frameworks for live inference pipelines and mentored the software engineering team on robust production ML practices.
- Cut search complaints and negative feedback by 12% and raised search engagement by 16% by deploying custom Elasticsearch analyzers, strategic field boosting, and relevance tuning.

Data Scientist, *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Optimized an XGBoost GLR model, driving a 3% relative increase in AUC-ROC over the legacy model, validated via rigorous A/B testing, and utilized SHAP values to identify key demographic drivers (income, age, household composition).
- Deployed uplift modeling to isolate creative-driven lifts in GLR, and validated out-of-sample models as a leading indicator of member rate 60 days in advance ($R^2 = 0.28$).
- Executed model-driven targeting and creative iterations that boosted overall GLR by 18% in relative terms.
- Enhanced geospatial targeting precision by 20% using GeoPandas for advanced GIS spatial analysis, effectively shifting marketing resources to high-opportunity regions.
- Orchestrated the monthly production monitoring of up to 10 live client models (tracking accuracy, feature drift, and forecast-vs-actual variances) and engineered a company-wide self-service dashboard built on curated Snowflake and Databricks data pipelines.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes with tenure-dependent churn (10%, 5%, then 2% monthly) to show the LTV = ARPU / churn identity holds only under constant churn.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), matching the simulated mean.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members than a prior-year-cost baseline.

EDUCATION

Ph.D. in Mathematics, *University of Pittsburgh* 2017 - 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics, *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Causal Inference & Experimentation: A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Observational Causal Methods, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs), Robustness Checks

Modeling & Statistics: Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, SHAP, Drift Monitoring

Data & Tools: SQL, Python (pandas, statsmodels, scikit-learn, econml), R (survey analysis), C#, Snowflake, Databricks, Spark, dbt, Curated Tables & Data Marts, AWS, Azure, Power BI, Git

Collaboration: Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring

Metrics & Planning: KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Consumer Product Analytics (Acquisition, Activation, Engagement, Retention, Monetization), CAC, LTV / ARPU, Contribution Margin, Scenario Analysis, Target Setting, Budget Allocation, Forecasting